Case Studies in Construction Materials 22 (2025) e04254

Contents lists available at ScienceDirect

Case Studies in Construction Materials

journal homepage: www.elsevier.com/locate/cscm

Research on 3D printing concrete mechanical properties prediction model based on machine learning

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ARTICLE INFO

Keywords:
3D printing concrete
Machine learning
Mechanical properties prediction
Random forest model
Data preprocessing

ABSTRACT

This study proposes an effective machine learning-based prediction method to satisfy the urgent requirement to anticipate the mechanical properties of 3D-printed concrete. The goal is to support the accurate use of 3D printing technology in the building sector. We have successfully created machine learning models that can predict compressive strength and flexural strength by combining experimental data from a variety of 3D printed concrete samples and carefully preparing the data. Our study explores the fundamentals and practicality of several models, such as artificial neural networks, decision trees, random forests, support vector regression, and linear regression. We have made sure that our prediction findings are reliable and scientifically sound by implementing stringent model training and validation procedures. With a correlation coefficient between 0.96 and 0.98 with real values, experimental results demonstrate the random forest model's remarkable predicted accuracy, greatly beyond that of conventional prediction techniques. The practical use of 3D printed concrete in engineering projects is strengthened by this work, which also opens up new avenues for investigation and highlights the enormous potential of machine learning to improve the prediction of mechanical properties of building materials.

1. Introduction

In recent years, the rapid advancements in technology have ushered in unprecedented opportunities and challenges for the application of 3D printing technology in the architectural sector [1–3]. Notably, 3D printing of concrete, with its inherent attributes of efficiency, flexibility, and environmental friendliness, is progressively emerging as a pivotal force driving innovation and transformation within the construction industry [4–6]. With the development of 3D printing technology in the field of construction, the study of the mechanical properties of 3D-printed concrete has become increasingly focal. Current research is not only committed to improving printing efficiency and buildability but also delving into the mechanical response of concrete during the printing process. Recent research advancements include numerical simulation and experimental testing of the early mechanical behavior of 3D-printed concrete, with detailed elaboration by R.J.M. et al., revealing the mechanical behavior of concrete during printing, which is crucial for understanding material response. Furthermore[7], Changbin Joh et al. have explored the key factors affecting the buildability and mechanical properties of 3D-printed concrete, providing a scientific basis for enhancing printing quality and structural strength[8]. Meanwhile, Huawei Liu et al. have proposed new methods for calculating the strength of 3D-printed concrete masonry wall structures

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https://doi.org/10.1016/j.cscm.2025.e04254
Received 8 October 2024; Received in revised form 1 January 2025; Accepted 13 January 2025
Available online 15 January 2025
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under various load conditions, which is significant for structural design and safety assessment[9]. Yanming Liu et al. have provided a comprehensive perspective, covering the preparation, performance, and ecological applications of planting concrete. The article suggests that the main factors influencing the design of planting concrete include the water-cement ratio, coarse aggregate size and gradation, the ratio of aggregate to cement, and porosity[10], which also serve as references for the input conditions affecting mechanical properties in this paper. H Huang et al. proposed a new type of replaceable artificial controllable plastic hinge (ACPH) for precast concrete beam-column joints, and explored the impact of ACPH on the seismic behavior of the joints. Through this study, a more comprehensive analysis of the seismic performance of 3D-printed concrete can be conducted[11]. These studies not only enrich our understanding of the mechanical behavior of 3D-printed concrete but also provide a solid scientific foundation for future applications. However, a major technological barrier preventing the widespread use of this technology is the precise prediction of mechanical properties, namely the evaluation of compressive and flexural strength, in 3D-printed concrete. This is mostly because 3D-printed concrete has special material properties and a unique molding technique. Even if they are trustworthy, traditional laboratory testing techniques have a number of drawbacks when used with 3D-printed concrete. First, the complexity of sample preparation is heightened due to the intricate geometric shapes and internal structures of 3D-printed concrete, which often exceed the precision and complexity that traditional methods can accurately replicate [12]. There may also be differences between the mechanical performance test findings of the printed specimens and the actual mechanical characteristics of the material because of the anisotropy caused by the printing direction, which also results in anisotropic mechanical properties. Second, the time-consuming nature of traditional tests, such as compression and flexural tests, is unsuited for the rapid feedback and iterative design processes required by 3D-printed concrete technology. These methods typically require extensive periods for sample preparation, testing, and data analysis, presenting a notable bottleneck in an industry that demands swift responsiveness. Third, the high cost associated with the preparation and testing of 3D-printed concrete samples, coupled with the need for specialized equipment and labor, makes traditional testing methods relatively expensive. This costliness limits the breadth and depth of research, particularly in research environments with constrained resources. Lastly, traditional laboratory methods often focus on a single aspect of performance, such as mechanical strength, and struggle to comprehensively assess the multifaceted performance of 3D-printed concrete, which includes not only strength but also durability, permeability, and other critical factors. This limitation hinders a holistic evaluation of the material's suitability for various applications. Consequently, the quest for a rapid, accurate, and cost-effective prediction methodology has become increasingly imperative. The need for an approach that can address these challenges and provide a more comprehensive understanding of 3D-printed concrete's performance is clear. It is against this backdrop that our study aims to develop a machine learning-based prediction model that can overcome the limitations of traditional testing methods and meet the demands of the modern construction industry for swift and precise control.

Against this backdrop, the emergence of machine learning (ML)[13] technology offers a novel perspective and toolset for addressing the aforementioned challenges. Leveraging its formidable data processing capabilities and pattern recognition abilities, machine learning has demonstrated remarkable predictive performance across various domains[14,15]. Significant progress has been made in the application of machine learning technology within the field of 3D printing, particularly in the prediction of material properties. For instance, Goh, G. D. et al. optimized the laser powder bed fusion process for Ti6Al4V through a data imputation strategy [16], while Cao, Y. utilized machine learning to aid in the prediction and optimization of the mechanical properties of Ti6Al4V alloy [17]. These studies demonstrate that machine learning technology can effectively process and analyze complex data generated during the 3D printing process, offering new perspectives for material property prediction and process optimization. Within the realm of materials science, and particularly in the prediction of material properties, machine learning has achieved notable successes. Machine learning has become increasingly integral to civil engineering, with applications ranging from predicting material properties and optimizing material preparation to infrastructure health monitoring, construction site surveillance, and smart city development. Y Niu et al. proposed a deep learning model based on convolutional neural networks, which was trained with data from finite element models and effectively predicted the stress and damage of concrete under various working conditions[18]. L He et al. established a new theoretical model for the probability distribution of pore length in 3D-printed concrete based on the simulation of zigzag interfaces, predicted and verified the quasi-exponential distribution of pore length, and studied the relationship between interlayer strength, porosity, and the number of pores[19]. D Lu et al. proposed a fractional-order elastoplastic constitutive model that can more accurately capture the nonlinear mechanical behavior of concrete under complex stress states[20]. De-Cheng Feng et al. introduced an Adaptive boosting algorithm to forecast the compressive strength of concrete, harnessing the power of multiple weak learners to construct a strong predictor that maps input data to output responses[21]. Zijian Wang et al. developed an imaging algorithm using interfacial waves and transducer array technology to detect cracks in underwater concrete structures, with an absolute error in crack localization of less than 7 millimeters, thereby validating the reliability of machine learning models in predicting mechanical properties[22]. Zhang J et al. addressed the optimization of concrete mixtures by proposing a multi-objective optimization (MOO) approach that integrates machine learning with metaheuristic algorithms, aiming to optimize multiple objectives such as strength, cost, and slump under highly nonlinear constraints. In the realm of infrastructure health monitoring[23], Ai L presented an automated damage localization method based on a single acoustic emission sensor, utilizing data fusion and deep learning algorithms for precise identification and localization of damage on large steel plate specimens simulating nuclear waste storage tanks[24]. Laxman Kc et al. developed a comprehensive automated framework for crack detection and depth assessment in concrete structures using images captured by portable devices, employing a binary Convolutional Neural Network (CNN) model for crack detection and combining CNN feature extraction layers with regression models (Random Forest and XGBoost) for predicting crack depth, which has been proven to be accurate and reliable for assessing the condition of concrete structures and selecting appropriate repair methods. Construction site monitoring has also benefited from machine learning advancements[25], with Kim J proposing a semantic graph neural network approach that represents construction resources and their complex interactions through graph structures and simulates propagation

effects using neural information transfer mechanisms, achieving precise detection of entangled visual relationships in actual construction images and improving the F-score by 11.5 % compared to baseline models[26]. Cheng JP introduced a framework that combines worker re-identification (ReID) and personal protective equipment (PPE) classification, addressing these two challenges with deep learning techniques to enable comprehensive analysis of worker behavior across multiple cameras and enhancing the accuracy of worker ReID and PPE classification by 4 % and 13 %, respectively. This contributes to video analysis and safety compliance checks on construction sites[27]. Lastly, the integration of machine learning into smart city initiatives and digital twin technology is paving the way for more intelligent and sustainable urban development. However, its application in predicting the mechanical properties of 3D-printed concrete remains in its infancy[28]. Consequently, the objective of this study is to construct a precise prediction model tailored specifically for the mechanical properties of 3D-printed concrete through the deep integration of machine learning technology. This endeavor aims to fill the research gap and contribute to advancing this nascent field[15].

The rationale for selecting machine learning as the solution approach in this study primarily stems from the following considerations: Firstly, machine learning algorithms excel at handling complex, nonlinear relationships, enabling them to discern implicit patterns and laws within data. This is crucial for comprehending and predicting the intricate interactions within the mechanical properties of 3D-printed concrete. In the study of the mechanical behavior of 3D-printed concrete, we have observed that its behavior often involves complex nonlinear relationships. To more accurately describe these relationships, we have introduced the theoretical foundation of nonlinear strength criteria. Nonlinear strength criteria, Dechun Lu emphasize the impact of strain rate on the dynamic strength of concrete and propose a nonlinear dynamic uniaxial strength criterion based on physical mechanisms[29]. This theoretical foundation not only describes the actual dynamic strength but also reflects the ultimate strength of concrete under high strain rates. In our research, by introducing a similar nonlinear theoretical framework, we aim to gain a deeper understanding and predict the mechanical response of 3D-printed concrete under various loading conditions. Secondly, machine learning models can effectively mitigate the reliance on extensive experimental testing, thereby enhancing prediction efficiency and cost-effectiveness. Lastly, through continuous optimization and training, these models can progressively elevate prediction accuracy, providing a robust foundation for reliable application in practical scenarios[30,31]. The mechanical properties of 3D-printed concrete are influenced by a multitude of factors and exhibit complex nonlinear relationships. Traditional empirical models struggle to accurately capture these intricate relationships, whereas machine learning models, especially non-parametric ones, are more adept at handling such complexity. The mix design for 3D-printed concrete is a challenging and iterative task that typically requires a significant amount of experimentation and testing to achieve an optimal design. Machine learning provides an advanced analytical technique that can more efficiently and accurately determine the concrete mix ratios with appropriate rheological and mechanical properties , effectively reducing the reliance on the "trial batch" method.

In reviewing past research, we have observed that while notable advancements have been made in areas such as 3D printed concrete's preparation techniques, material mix optimization, and fundamental mechanical property testing[32,33], studies focused on mechanical property prediction remain fragmented and lack systematic organization. Particularly, the application of machine learning (ML) techniques to predict the mechanical properties of 3D printed concrete is scarce. Existing research primarily concentrates on the utilization of specific algorithms, with inadequate comprehensive comparisons and optimized selection among multiple algorithms. Furthermore, the relatively small scale of datasets employed fails to comprehensively capture the intricate characteristics of 3D printed concrete. Regarding the preparation techniques and material mix optimization of 3D printed concrete, Dinglue Wu pioneered the introduction of a novel ML model based on a stacking strategy to predict the interlayer bonding strength (IBS) of 3D printed concrete[34]. This reflects a growing trend among researchers to leverage machine learning models in optimizing performance predictions during the optimization of preparation techniques and material ratios. Xiaofei Yao et al. developed two machine learning (ML) models in conjunction with the Beetle Antennae Search (BAS) technique to achieve predictive capabilities for the mechanical properties of steam-cured concrete. [35], This endeavor represents a significant attempt to optimize the material composition and preparation process. In the realm of basic mechanical property testing and prediction for 3D-printed concrete, Sohaib Nazar et al. have employed decision trees (DT) and Bagging regressors (BR) to accurately forecast the plastic viscosity (PV) and yield stress (YS) of concrete with varying mix proportions. This predictive approach constitutes a pivotal aspect of fundamental mechanical property assessment [36]. Ronghua Zhu et al. have devised a machine learning prediction model utilizing both experimental and numerical outcomes. This model aims to accurately forecast the compressive and flexural strengths of 3D-printed concrete reinforced with aggregate mixtures, thereby directly addressing the pivotal mechanical properties that lie at the heart of 3D printed concrete research [37]. Ammar Ali et al. have developed multiple predictive models, including Gaussian Process Regression and Decision Tree Regression, to anticipate the bending and tensile strengths of 3D printed concrete, thereby also emphasizing the prediction of mechanical properties[38]. Hamed Izadgoshasb et al. have integrated optimization algorithms with mechanical property prediction by employing a multi-objective grasshopper optimization algorithm in conjunction with an artificial neural network to calculate and forecast the compressive strength of extruded 3DP concrete[39]。Mana Alyami et al. showcased the versatility of various machine learning algorithms by utilizing seven distinct algorithms to predict the compressive strength of 3D-printed fiber-reinforced concrete, highlighting their application in performance prediction[40].Alireza Ghasemi accurately predicted compressive strength performance through interpretable artificial intelligence algorithms such as multiple linear regression, random forest, and XGBoost, emphasizing the explainability of these algorithms[41].In exploring the unique properties and applications of 3D-printed concrete, Wenxin Lao leveraged machine learning techniques and an artificial neural network model to correlate the formation of extrudates with printing parameters, investigating the influence of printing parameters on extrudate formation[42].Song-Yuan Geng et al. introduced a multi-objective optimization theoretical framework grounded in machine learning algorithms for the design of 3D-printed low-carbon concrete, addressing the sustainability requirements of the construction industry by harmonizing environmental protection with performance optimization[43]. Xianlin Wang utilized deep generative adversarial network-enhanced interpretable ensemble learning

to predict the rebar bond performance in 3D concrete printing, focusing on the structural performance of 3D-printed concrete[44].

Through comparative analysis, it becomes evident that the integration of 3D printing concrete technology with machine learning not only enhances prediction accuracy and efficiency but also provides scientific guidance for the further optimization and application of 3D printed concrete technology [12,14,32]. As a result, the current study expands on the strong framework established by earlier investigations. We hope to make significant progress in forecasting the mechanical characteristics of 3D-printed concrete by increasing the size of the dataset, incorporating a wide range of representative machine learning models, and methodically training and evaluating these models. Beyond merely increasing forecast efficiency and accuracy, we hope that our research will provide technical assistance and scientific justification for the further development and broad use of 3D-printed concrete technology. Furthermore, we hope to encourage further research and broad use of machine learning in the field of forecasting the characteristics of construction materials.

2. Data and models

2.1. Data collection and selection

The 3D printing concrete technology exhibits significant differences in material requirements compared to traditional concrete, placing particular emphasis on the precise selection and scientific proportioning of cementitious materials, aggregates, mineral admixtures, and additives[45]. The mechanical characteristics and workability of 3D printed concrete are significantly influenced by the special qualities of these materials. Thus, this study offers a thorough explanation of the properties of the 3D printed concrete that was utilized, going into the reasoning behind the selection of different kinds of materials and their distinct contributions to the overall performance in addition to elucidating the specifics of its proportioning.

In order to guarantee data quality and applicability, this study followed three fundamental standards for data screening: representativeness, consistency, and completeness. In order to ensure the thoroughness and correctness of the study analysis, the completeness principle gave top priority to the selection of data records that included full mechanical property test results, especially those that included both compressive and flexural strength testing. By guaranteeing that all data were collected under comparable or equivalent testing conditions, the principle of consistency sought to exclude any potential influence of outside factors on the study's findings. The principle of representativeness sought to comprehensively reveal the performance characteristics of 3D-printed concrete by meticulously selecting data samples that represented diverse mix proportions, printing parameter settings, and curing conditions, thereby enhancing the general applicability and insightfulness of the research conclusions. When selecting data, we paid particular attention to the following points:

(1) Data Comprehensiveness and Diversity: We prioritized data records that provided comprehensive mechanical property test results, particularly those that included both compressive and flexural strength tests. This approach ensures that the model can learn mechanical properties under various conditions, enhancing its generalization capabilities to adapt to different practical application scenarios. Our dataset encompasses a diverse range of 3D-printed concrete mixtures, printing parameters, and curing conditions, which is crucial for capturing the breadth of possible outcomes in 3D-printed concrete.
(2) Data Quality: High-quality data is crucial for the success of machine learning models, as any bias or error can lead to inaccurate predictions. Therefore, we prioritized samples with complete and highly accurate records, ensuring that the data used in our analysis is reliable and trustworthy.
(3) Data Representativeness: To ensure our data represents the typical performance of 3D-printed concrete, we selected samples covering various strength levels and component ratios. This reflects the actual range of 3D-printed concrete applications and provides a more accurate basis for our models to make predictions.
(4) Data Availability: Considering the limited availability of open-source datasets, we supplemented our dataset with data from our own experiments and literature collection, increasing the sample size and improving the model's predictive performance. This strategy ensures a broad and diverse range of data, effectively training and testing our models.

Through a meticulous and rigorous screening process, a diverse and comprehensive experimental dataset encompassing nearly 500 sets of data was meticulously compiled, forming a robust foundation for the thorough conduct of this study. Specifically, we have incorporated 425 sets of data from our laboratory experiments, which were conducted under stringent and controlled conditions to ensure the highest level of accuracy and consistency. These data were collected over a period of 36 months, with each experiment repeated at least three times to ensure reliability. Additionally, we have included 175 sets of data from published literature that met our strict inclusion criteria, ensuring a broad and diverse range of data to train and test our models effectively. It is important to note that cross-validation techniques were employed, where applicable, to mitigate any potential bias arising from the inclusion of a substantial portion of data from a single source. Specifically, our laboratory data was not used to exclusively train models but was integrated with external data to validate and generalize our findings, thereby enhancing the credibility and applicability of the results.

Table 1 comprehensively covers the fundamental properties of 3D-printed concrete materials, encompassing not only the physical and chemical characteristics of key components such as cement, aggregates, admixtures, fibers, water-reducing agents, and viscosity modifiers (with a three-level classification and attribute description). Water to binder ratio (W/C) is the mass ratio of water to binding materials in concrete. W/C affects the fluidity and mechanical properties of concrete. Cement is the primary binding material in concrete, which is hardened by hydration reaction after water mixing, which plays a major role in the mechanical properties of concrete. Ordinary Portland cement (P.O.) is the most commonly used cement, and the main component is silicate. Composite Portland

Table 1
Dataset parameters and symbols.

Class 1	Class 2	Class 3	Types and grades	Influence	Symbol	Operation
W/B				Mass ratio of water to cementitious material in 3D printed concrete	I1	Input
Cement	**P O**		42.5	Ordinary Portland cement makes a major contribution to mechanical properties through a hydration reaction	I2	Input
			42.5 (mrket-hailuo)		I3	Input
			42.5 (market-aota)		I4	Input
			42.5 (internet-hailuo)		I5	Input
	P C		42.5 (market-jinyu)	Similar to P.O	I6	Input
	Double fast	**SAC**	42.5	Cement with faster hydration reaction rate and early mechanical properties	I7	Input
		Belite			I8	Input
Mineral admixture	**Fly ash**		Level 1	It has a certain hydration reactivity, which is used to replace part of the cement and reduce the water-glue ratio.	I9	Input
			Level 2		I10	Input
			Level 3		I11	Input
	Blast		S95	It has a certain hydration reactivity, which is used to replace part of the cement	I12	Input
	Coal Gangue		< 0.08 mm	It mainly plays a role in filling and thickening, has a relatively high hydration reactivity	I13	Input
	Cinder		< 0.08 mm		I14	Input
	RWTB		water demand ratio		I15	Input
	Powder		110 %			
	Stone		0.063–0.15 mm		I16	Input
	Powder		0.01–0.04 mm		I17	Input
	Silica Fume			Ultra-fine silica powder has a good filling effect and hydration reactivity and has a certain thickening effect	I18	Input
Aggregate	**Natural sand**	**River sand**	fine	General construction sand	I19	Input
			medium		I20	Input
		Sand (yellow river)			I21	Input
		Quartz sand	0.3mm−0.6 mm		I22	Input
			0.15mm−0.3 mm		I23	Input
		Standard sand			I24	Input
	Solid waste sand	**Coal Gangue (zaozhuang)**	0–5 mm	Coal mine by-product, with lower strength and density	I25	Input
			0.16–5 mm		I26	Input
		Coal Gangue (gansu)	1.25–0.315 mm		I27	Input
			0.63–0.16 mm		I28	Input
			0.315–0.08 mm		I29	Input
			2.5mm−0.315 mm		I30	Input
		Slag (gansu)	0.315mm−0.08 mm	Bottom slag of coal-fired power plants, which has a lower density and strength	I31	Input
		Iron tailings	wash	Magnetite by-products, similar to construction sands	I32	Input
		Copper tailings	0	Fine powder, high water demand	I33	Input
	lightweight	**Cenosphere**		Hollow microspheres	I34	Input
Admixture	**Water glass**		M= 1.5	sodium silicate	I35	Input
	Water-reducer	**FDN**		Improve the fluidity of concrete and reduce W/C	I36	Input
		PCE	Sika		I37	Input
	HPMC		25000mpa·s	It has the effect of thickening and water retention	I38	Input
	Air Entraining	**SDS**		During the mixing process, micro air bubbles are introduced into the concrete	I39	Input
	UEA			U-type expansive agent for concrete,	I40	Input
	Defomer			Eliminate oversized air bubbles	I41	Input
	Thixotropic	**Fumed silica**		Ultra-fine silica powder has a good filling effect and hydration reactivity and has a certain thickening effect	I42	Input
		Calcium based bentonite		It plays a role in lubrication and thickening	I43	Input
Fiber		**RWTB**	3–6 mm	The main components are resin and glass fiber	I44	Input
		Basalt	6 mm		I45	Input
			15 mm		I46	Input
		Glass	18 mm		I47	Input
		PP	18 mm		I48	Input
			12 mm		I49	Input
			6 mm		I50	Input
Mechanical strength	**3d**	**Flexural strength**			O1	Output
		Compressive strength			O2	Output
	7d	**Flexural strength**			O3	Output

(*continued on next page*)

Table 1 (*continued*)

Class 1	Class 2	Class 3	Types and grades	Influence	Symbol	Operation
		Compressive strength			O4	Output
	28d	**Flexural strength**			O5	Output
		Compressive strength			O6	Output

cement (P.C.) adds a small amount of slag, ash and other raw materials, and its performance is similar to that of P.O. Sulphate aluminate cement (SAC) is a cement with calcium sulfoaluminate as the main component, which has a faster hydration reaction rate and higher early mechanical properties than Portland cement. Mineral admixtures are a type of mineral powder, usually derived from solid waste, and are used to replace part of the cement in concrete, as well as to enhance or adjust the performance of concrete. Aggregates contribute to the majority of the concrete's volume and play a crucial role in determining the material's strength. The type, size, and grading of aggregates can impact the mechanical properties of concrete. Admixtures are added to modify the properties of concrete, such as workability, setting time, and durability. They can enhance the performance of concrete and are essential for adjusting the mix proportions according to specific printing requirements. Fibers are typically included to improve the toughness and crack resistance of concrete. They contribute to the overall mechanical behavior, especially in terms of flexural strength and impact resistance. Water-reducing agents reduce the water demand in the mixture while maintaining workability, which is crucial for the printability of 3D-printed concrete. They help achieve the desired flow and shape during the printing process. Given the specific challenges of extrusion-based 3D printing, viscosity modifiers are used to control the consistency of the concrete mixture. This ensures that the material can be accurately extruded and maintains its shape after printing. These parameters are vital to the 3D printing process and directly affect the quality and strength of the printed concrete. By including them as input variables, the mechanical

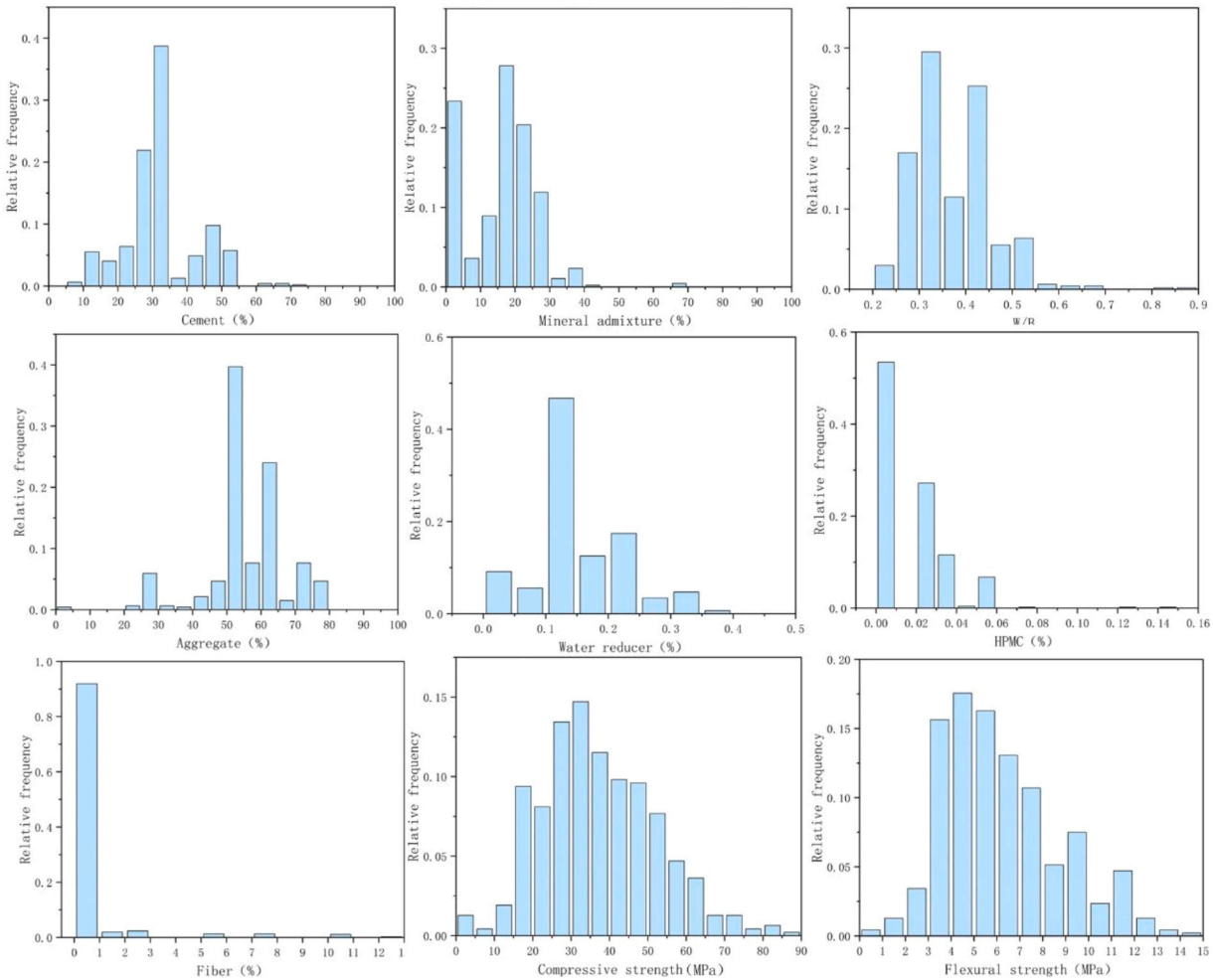


Fig. 1. Histograms of characteristics of the dataset.

properties of 3D-printed concrete structures can be better predicted and controlled. The dataset utilized in this study exhibits a diverse range of variables, highlighting the robustness of the research. Fig. 1 provides an illustration of the feature distribution within the dataset.

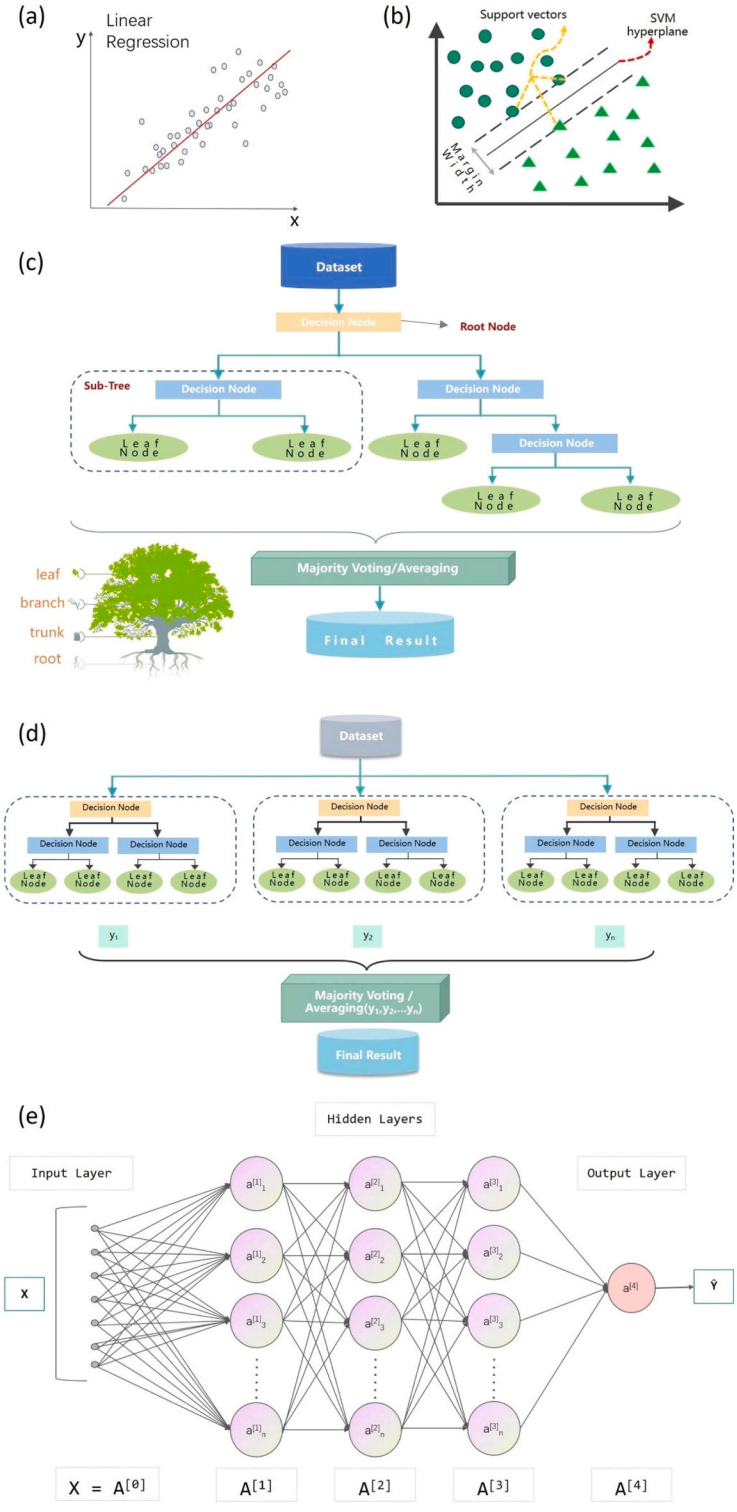


Fig. 2. General Schematic Diagram of Machine Learning Models. (a)LR, (b)SVR, (c)DT, (d)RF, (e)ANN.

2.2. Data cleaning and preprocessing

During the data cleaning step, this study prioritized removing missing values, outliers, and duplicate entries to assure data quality. For missing values, appropriate interpolation methods were used, and reasonable estimates were generated using contextual information. Outliers were thoroughly examined, with modifications or removals made based on their real circumstances. Duplicate records were merged or eliminated to ensure the dataset's accuracy and consistency. The study improved data quality through these pretreatment processes, laying the groundwork for future data analysis and model development.

2.3. Selection of machine learning models

In predicting the mechanical properties of 3D-printed concrete, the choice of an appropriate machine learning model is paramount. This study selected five representative machine learning models for in-depth exploration: linear regression, support vector regression, decision trees, random forests, and neural networks[46]. A more thorough and accurate forecast of the mechanical properties of 3D-printed concrete may be obtained thanks to the unique features that each of these models has that allow them to extract profound insights from data in several dimensions. Among them, multiple linear regression is a linear model that assumes a linear relationship between input features and output variables. However, this model has limitations when dealing with complex nonlinear relationships, as it cannot capture the interactions and nonlinear relationships between variables. In contrast, nonlinear models such as decision trees, random forests, and artificial neural networks show clear advantages in handling such complex relationships. Decision trees capture nonlinear relationships by recursively partitioning data, while random forests enhance the robustness and accuracy of predictions by integrating multiple decision trees. Artificial neural networks, with multiple hidden layers and nonlinear activation functions, can fit complex nonlinear relationships, making them excellent at capturing complex patterns in data. Support vector regression are models that can be either linear or nonlinear, depending on the kernel function used. Support vector regression maps data into high-dimensional space through the kernel function, thus finding the optimal linear separation plane in high-dimensional space, which makes support vector regression very effective in dealing with nonlinear problems. Although linear models have advantages in terms of simplicity, interpretability, and computational efficiency, they are less suitable when dealing with materials with complex nonlinear characteristics, such as 3D-printed concrete. Nonlinear models are more applicable because they can capture more complex data patterns. In our research, by comparing the performance of these models, we found that nonlinear models provide higher accuracy and flexibility in predicting the mechanical properties of 3D-printed concrete.

Linear Regression (LR)[47], as a fundamental and intuitive prediction tool, revolves around predicting the target variable by identifying the best-fit straight line, as depicted in Eq. (1) and Fig. 2 (a). In this study, we leveraged the proportions of raw materials and other experimental conditions as independent variables to forecast the compressive and flexural strengths of 3D-printed concrete. The choice of Linear Regression as a baseline model stems not only from its simplicity and clarity but also from its ability to directly reflect the linear dependency between independent and dependent variables. While the mechanical properties of 3D-printed concrete may be influenced by numerous nonlinear factors, the linear model serves as a starting point, facilitating an initial understanding of the fundamental trends within the data.

$$y = \beta_0 + \beta_1 \quad x_1 \quad + \beta_2 \quad x_2 \quad + \bullet\bullet\bullet + \beta_n \quad x_n \quad + \varepsilon \# \tag{1}$$

Where β_0 , β_1,…, β_n are the model parameters, and ε represents the error term.

Support Vector Regression (SVR) [48]represents the extension of Support Vector Machines (SVM)[49] to regression problems. This model aims to identify an optimal hyperplane that minimizes the distance from all data points to this hyperplane, while introducing an epsilon-insensitive loss function (ϵ) to allow for a certain degree of error. A schematic diagram illustrating this concept is presented in Fig. 2 (b). Key concepts in SVR include support vectors, kernel functions, and soft margins. When dealing with high-dimensional data or complex nonlinear relationships, SVR excels. By seeking an optimal separating hyperplane, SVR ensures that the differences between predicted values and actual observations are minimized, enabling its outstanding performance in tasks such as concrete strength prediction.

Decision Tree (DT)[50,51] constructs a tree-like structure by recursively partitioning the data into subsets of greater purity. Each internal node represents a decision condition on an attribute, each branch signifies an outcome of that decision, and each leaf node corresponds to a class label or regression value. A schematic diagram illustrating this is shown in Fig. 2 (c). Key concepts include information gain, gain ratio, and Gini index, which are used to select the optimal attribute for splitting. When dealing with datasets containing multiple features, these methods have demonstrated their effectiveness in concrete strength prediction. In applying this algorithm, our study employed cross-validation techniques to select optimal model parameters, thereby mitigating the risk of overfitting and ensuring that the model maintains robust predictive performance when confronted with new data. Capable of addressing both classification and regression problems, decision trees hold extensive application potential in the field of material performance prediction.

Random Forest (RF) [52]is an ensemble learning model that integrates multiple decision trees. Each tree is trained on a randomly sampled subset of the training data, and during the construction of each tree, only a randomly selected subset of attributes is considered at each node split[53]. The final prediction is derived as the average of all trees' predictions for regression problems or the mode for classification problems. A schematic diagram illustrating this concept is presented in Fig. 2 (d). Key concepts include Bootstrap sampling, random attribute selection, and ensemble learning. This method effortlessly handles large numbers of features and excels in dealing with nonlinear relationships and feature selection. By leveraging ensemble learning, Random Forest enhances

prediction stability and accuracy, enabling it to process a vast array of input variables, automatically perform feature selection, and exhibit robustness against outliers and noisy data. These strengths render Random Forest particularly adept at tackling complex tasks such as predicting the mechanical properties of 3D-printed concrete.

Artificial Neural Network (ANN)[54,55] is an adaptive system that learns from the features within data to identify patterns, classify data, or make predictions. Composed of multiple layers, including an input layer, hidden layers (which may comprise multiple levels), and an output layer, this model incorporates neurons within each layer that communicate through weighted connections[56]. ANN operation relies on two core processes: forward propagation and backpropagation. Forward propagation entails the passage of input data through the network's hierarchical structure until an output is generated. Conversely, backpropagation adjusts the network weights based on the output error, achieving this by calculating error gradients and applying optimization algorithms such as gradient descent to update the weights, thereby reducing the error and enhancing model accuracy. A schematic diagram illustrating this is presented in Fig. 2**. (e)**. Key concepts in this domain encompass neurons, activation functions (e.g., Sigmoid, ReLU), the back-propagation algorithm, and gradient descent optimization. Characterized by adaptability, nonlinearity, and parallel processing, ANNs are widely applied across various domains, including pattern recognition, classification, regression, and prediction. In the context of 3D-printed concrete, ANNs provide a robust framework for understanding and predicting material behavior. By incorporating ANNs into our predictive models, we leverage their advanced learning capabilities to address the complex, multivariable nature of 3D-printed concrete's mechanical properties, thereby enhancing the accuracy and reliability of our predictions.

In order to facilitate the effective creation of data science and machine learning projects, we used PyCharm as the Integrated creation Environment (IDE) and Python 3.12 as the programming language in this study. We imported a number of essential Python packages in order to precisely forecast the mechanical characteristics of 3D-printed concrete. In particular, we used numpy to perform mathematical operations efficiently, especially for large-scale array and matrix operations. Pandas was employed for data cleaning, organization, and analysis, with its powerful DataFrame object significantly simplifying the data processing workflow. Matplotlib. pyplot facilitated data visualization, enabling the intuitive presentation of prediction results and model performance through charts and graphs. From the sklearn.model_selection module, we imported train_test_split for dataset partitioning and cross_val_score for cross-validation, thereby assessing the model's generalization ability. For constructing machine learning models, we sourced LinearRegression from sklearn.linear_model to implement the linear regression model; SVR (Support Vector Regression) from sklearn.svm to handle nonlinear relationships; DecisionTreeRegressor from sklearn.tree to establish the decision tree model; RandomForestRegressor from sklearn.ensemble to integrate multiple decision trees for enhanced prediction stability; and MLPRegressor

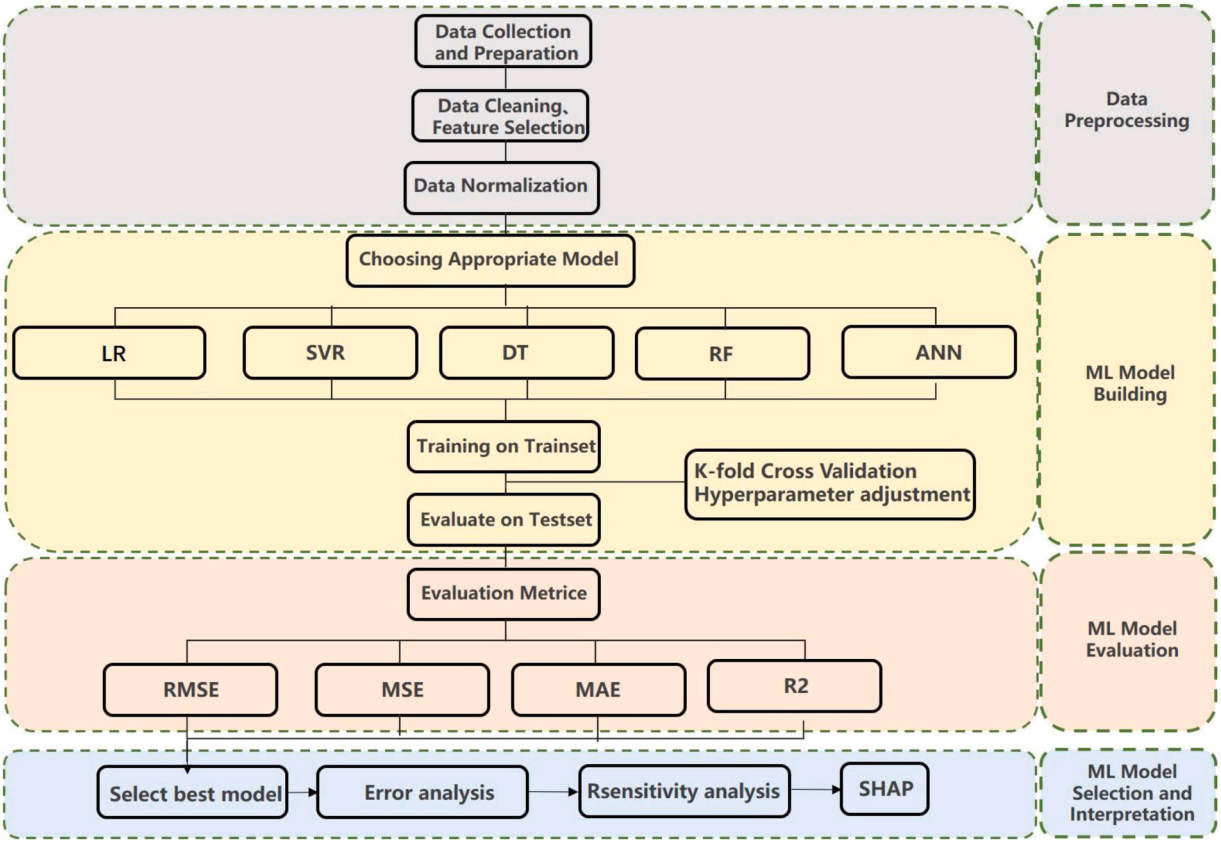


Fig. 3. technical roadmap.

(Multi-Layer Perceptron Regressor) from sklearn.neural_network to explore the application of neural networks in prediction tasks.

To enhance model performance, we introduced make_pipeline from sklearn.pipeline, which enables the encapsulation of multiple transformation steps and estimators into a single pipeline, thereby streamlining the model training and prediction process. Furthermore, StandardScaler from sklearn.preprocessing was employed for feature scaling to ensure numerical stability during model training. Lastly, we imported various evaluation metrics from the sklearn.metrics module, including mean_squared_error, mean_absolute_error, and r2_score (coefficient of determination R^2), to comprehensively assess the prediction accuracy and goodness-of-fit of the model.

3. Methodology

This chapter describes the process used to use machine learning models to forecast the performance of 3D-printed concrete, as seen in Fig. 3**.** To guarantee the accuracy and completeness of the dataset, data collecting and preparation are essential first steps. These include feature selection and data cleaning. Data normalization is then carried out to remove dimensional discrepancies and improve the models' resilience. After that, a suitable model—which could include LR, SVR, DT, RF, and ANN—is chosen in accordance with the particular issue at hand. The training set is then used to train these models, while the test set is used to assess them. During model training, K-fold cross-validation is employed to adjust hyperparameters and optimize model performance. Finally, various evaluation metrics such as Root Mean Square Error (RMSE), Mean Squared Error (MSE), Mean Absolute Error (MAE), and Coefficient of Determination (R^2) are utilized to assess model accuracy. Through these steps, the optimal model is selected, and subsequent error analysis, sensitivity analysis, and SHAP (SHapley Additive exPlanations) interpretations[57] are conducted to gain deeper insights into model performance and potential issues.

3.1. Data preprocessing

To ensure the accuracy and robustness of machine learning algorithms, this study performed meticulous preprocessing operations on the screened 500 sets of high-quality data prior to in-depth data analysis. These preprocessing steps are paramount as they significantly enhance data quality, ultimately optimizing the training effectiveness of the models. In addition to the contents mentioned in Section 2.2, to eliminate analytical biases arising from dimensional differences among various features, we normalized all feature data, mapping them within the range of [0,1]. This step aims to adjust all data to the same order of magnitude, ensuring that different features carry equal weight and influence within the machine learning models, thereby preserving the fairness and objectivity of the models.

The formula adopted for this step is (x - min_value) / (max_value - min_value + 0.001), where x represents a value in the original data; min_value is the minimum value in that feature column; and max_value is the maximum value in that feature column. The inclusion of 0.001 as a small constant in the denominator primarily serves to prevent division by zero when max_value equals min_value, ensuring the robustness of the code and avoiding division errors.

This study thoroughly combined correlation analysis methods with domain experience in terms of feature selection. We investigated the fundamental connections between important elements in concrete, including the water-binder ratio, aggregate makeup, and admixtures, and how these affect the material's compressive and flexural strengths. We discovered through recurrent trials that the cement's addition affects the concrete's fluidity and setting time. As a result, we examined how various cement varieties affected the mechanical characteristics of concrete that was 3D printed. The strength and stability of 3D-printed concrete are significantly influenced by the particles' homogeneity, size, and impurity content. Admixtures and additives are typically used to enhance the workability and durability of concrete, and we explored how these materials synergistically contribute to improving the performance of 3D-printed concrete. The addition of fibers reinforces the toughness and crack resistance of concrete. Furthermore, the water-cement ratio significantly affects the workability and strength of concrete. Additionally, printing process parameters such as printing speed and

Table 2
Pearson correlation coefficients of input parameters.

Symbol	CS-PCC	FS-PCC	Symbol	CS-PCC	FS-PCC	Symbol	CS-PCC	FS-PCC	Symbol	CS-PCC	FS-PCC
I1	0.904	0.714	**I14**	0.03	0.012	**I27**	0.039	0.029	**I39**	0.079	0.077
I2	0.043	0.04	**I15**	0.344	0.176	**I28**	0.043	0.028	**I40**	0.251	0.181
I3	0.362	0.326	**I16**	0.339	0.259	**I29**	0.041	0.029	**I41**	0.327	0.214
I4	0.248	0.138	**I17**	0.146	0.108	**I30**	0.04	0.016	**I42**	0.088	0.110
I5	0.152	0.064	**I18**	0.348	0.36	**I31**	0.036	0.015	**I43**	0.086	0.087
I6	0.322	0.19	**I19**	0.568	0.438	**I32**	0.546	0.443	**I44**	0.236	0.139
I7	0.228	0.212	**I20**	0.145	0.101	**I33**	0.4	0.347	**I45**	0.263	0.173
I8	0.474	0.390	**I21**	0.278	0.191	**I34**	0.207	0.109	**I46**	0.27	0.199
I9	0.632	0.463	**I22**	0.125	0.057	**I35**	0	0	**I47**	0.115	0.115
I10	0.213	0.139	**I23**	0.280	0.219	**I36**	0.485	0.421	**I48**	0.111	0.096
I11	0.083	0.077	**I24**	0.145	0.168	**I37**	0.753	0.587	**I49**	0.288	0.207
I12	0.526	0.365	**I25**	0.042	0.047	**I38**	0.573	0.440	**I50**	Nan	Nan
I13	0.033	0.03	**I26**	0.046	0.04						

The abbreviation "CS-PCC" stands for "Compressive Strength Pearson Correlation Coefficient ; the abbreviation "FS-PCC" represents "Flexural Strength Pearson Correlation Coefficient"

interlayer interval time also influence the mechanical properties of the final product. Given the limited data records available in existing literature regarding printing process parameters, which are insufficient to support the data volume required for machine learning, this study prioritized raw material data, abundant in both literature and our own experimental data, as the original input for machine learning. To enhance the scientific rigor and precision of feature selection, we employed advanced correlation analysis techniques to ensure that the selected features had a significant impact on the compressive and flexural strengths of concrete. The Pearson correlation coefficients are presented in Table 2. By analyzing the correlation between each feature and concrete strength, we screened out the most influential features for strength prediction and used them to build our models.

3.2. Dataset partitioning

To enhance the generalization ability of the models and ensure high-precision predictions, the entire dataset was partitioned into a training set and a test set at a ratio of 80 % and 20 %, respectively. Following numerous attempts with varying partition ratios, including 90 % and 10 %, 85 % and 15 %, 75 % and 25 %, as well as 70 % and 30 %, the combination of 80 % and 20 % was ultimately selected based on its demonstration of the highest correlation. The reason for choosing to use 80 % of the data as the training set is that it provides more information for the model and yields the highest predictive R^2 values, thereby better learning and generalizing the patterns in the data. Although a 70 % training set and 30 % test set is a common split ratio, we believe that using more data for training in this study allows for more efficient use of limited data resources, especially in the field of 3D printed concrete, where obtaining high-quality data is relatively challenging. The training set was utilized for the initial training of the models and the fine-tuning of parameters, ensuring that the models could capture the essential characteristics of the data. Conversely, the test set was exclusively designated to evaluate the predictive performance of the trained models, thereby ensuring their reliability in practical applications. This data partitioning approach not only necessitates exceptional performance on the training data but, more crucially, mandates outstanding predictive accuracy when confronted with unknown data.

3.3. Model training

(1) LR

The model parameters were estimated using the Least Squares Method to identify the best-fit line (or a hyperplane in multidimensional space) that minimizes the sum of squared residuals between predicted and actual values. Specifically, the goal was to find the parameter values that minimize this sum of squared residuals. The preprocessed data were input into the model, and the model parameters (slope and intercept) were adjusted through iterative optimization algorithms to minimize prediction errors. The key parameter settings for Linear Regression are presented in Table 4.

(2) SVR

SVR aims to find an optimal hyperplane in a high-dimensional space that separates data points of different classes. During training, the optimal hyperplane is identified by solving a convex quadratic programming problem based on the selected kernel function and tuned parameters. This process can be facilitated using the SVM module within the machine learning library scikit-learn. The key parameter settings for SVR are outlined in Table 4.

(3) DT

Decision Tree is a classification or regression method based on a tree structure. It recursively partitions the data feature space to construct a series of decision rules, ultimately forming a tree-like structure. In a Decision Tree, each internal node represents a feature attribute test, each branch signifies an outcome of that test, and each leaf node corresponds to a class label or prediction result. Commonly used Decision Tree algorithms include ID3, C4.5, CART, among others. In this study, we will select the CART (Classification and Regression Trees) Decision Tree algorithm based on the characteristics of the data and prediction objectives. The key parameter settings for the Decision Tree are presented in Table 4. The training process commences at the root node, where the optimal splitting feature is chosen to partition the data into child nodes. This process is repeated until the stopping criteria are met (e.g., when the number of samples in a node falls below a preset threshold or when all samples within a node share the same output value).

(4) RF

Multiple Decision Trees are constructed, each independently predicting the data. The output of Random Forest is the average of the predictions from all trees. By randomly sampling and selecting features, various distinct training sets are created. A separate Decision Tree is trained on each training set. Upon completion of training, a set of Decision Tree models is obtained. The key parameter settings for Random Forest are outlined in Table 4.

Table 3
MSE output values from 5-fold cross-validation.

Type	LR Cross-Validation MSE	SVR Cross-Validation MSE	DT Cross-Validation MSE	FE Cross-Validation MSE	ANN Cross-Validation MSE
Flexural Strength	1746369411.3856754	0.029953222401717865	0.09334676597400851	0.0021615584154548	0.020815800836208303
Compressive Strength	415800626.078904	0.08945907946667396	0.06386340339419383	0.04596439947687867	0.5506517270336861

Table 4
Model Hyperparameters.

Model	Parameters and Values	Annotation
"Linear Regression":LinearRegression	fit_intercept=True	# Calculating Intercept
"Support Vector Machine": SVR	C= 10 kernel= 'rbf'	# Regularization Parameter
	degree= 3	# Employing Radial Basis Function (RBF) Kernel
	gamma= 'scale'	# Dimension for Polynomial Kernel
	epsilon= 0.1	# Kernel Coefficient
		#Loss-Free Range within the Epsilon-Tube
"Decision Tree":DecisionTreeRegressor	criterion= 'squared_error'	# Employing Mean Squared Error (MSE)
	splitter= 'best'	# Selection of Optimal Split Point
	max_depth=None	# No Restriction on Tree Depth
	min_samples_split= 2	# Minimum Number of Samples Required to Split an Internal Node
	min_samples_leaf= 1	# Minimum Number of Samples Required to Be at a Leaf Node
"Random Forest":RandomForestRegressor	n_estimators= 100	# Number of Trees in the Forest
	criterion= 'squared_error'	# Utilizing Mean Squared Error (MSE)
	max_depth=None	# No Restriction on Tree Depth
	min_samples_split= 2	# Minimum Number of Samples Required for Splitting an Internal Node
	min_samples_leaf= 1	# Minimum Number of Samples Required for a Leaf Node
"Neural Network": MLPRegressor	hidden_layer_sizes= (100,)	# Single Hidden Layer with 100 Neurons
	activation= 'relu'	# Adoption of ReLU Activation Function
	solver= 'adam'	# Implementation of Adam Optimization Algorithm
	alpha= 0.0001	# L2 Regularization Parameter
	max_iter= 1000	# Maximum Number of Iterations

(5) ANN

A multi-layer neural network architecture is constructed, comprising an input layer, hidden layers, and an output layer. Each node (neuron) performs a nonlinear transformation on the input through an activation function and connects via weights and biases. The training process employs the backpropagation algorithm and a gradient descent optimizer to adjust the network's weights and biases. Through multiple iterations (i.e., epochs), the network gradually learns the mapping relationship from input features to output intensities. During training, a validation set can be utilized to monitor model performance and prevent overfitting. The key parameter settings for the neural network are presented in Table 4.

(6) 5-Fold Cross-Validation

In machine learning, cross-validation is a crucial method for evaluating model performance[39,54,58]. Cross-Validation helps prevent overfitting and underfitting by training and testing the model multiple times on different training and validation sets, effectively evaluating the model's performance on unseen data. This process aids in preventing overfitting and underfitting of the model; Cross-validation enhances the model's generalization ability by dividing the dataset multiple times and training and testing on different training and validation sets, helping us to determine if the model has overfitting or underfitting issues, thereby improving the model's generalization ability; Cross-validation makes full use of the data, ensuring that each data point is utilized when the sample

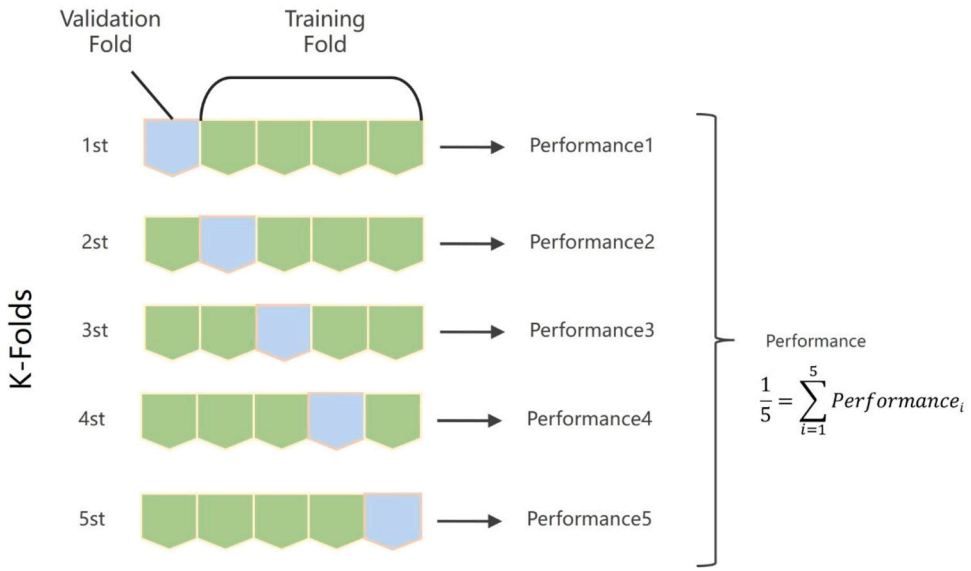


Fig. 4. Schematic Diagram of 5-Fold Cross-Validation.

size is limited, avoiding the waste of data that can occur when a portion of the data is fixed as a test set. In this study, we adopted 5-fold cross-validation, as illustrated in Fig. 4**.** Specifically, the entire dataset was evenly divided into five folds. During each iteration, four folds were used as the training set, while the remaining fold served as the test set. This process was repeated five times, with a different test set selected each time. The average evaluation results of the model performance, presented in Table 3**,** were obtained. The optimal parameters for the five models were determined through this 5-fold cross-validation approach. After parameter adjustment, the 5-fold cross-validation mean squared error (MSE) values for the SVR, DT, RF, and ANN models were relatively low, reflecting the predictive accuracy, model stability, and excellent generalization ability. However, the LR model yielded a relatively high MSE value, indicating underfitting. This suggests that using LR to predict mechanical strengths may be overly simplistic, unable to capture the complex relationships within the data. Additionally, the influence of randomness, particularly in smaller datasets, can lead to varying results depending on the data partitioning. Consequently, the results of a single cross-validation run may not be entirely reliable. Employing 5-fold cross-validation enables a more comprehensive assessment of model performance across different data subsets, enhancing the reliability and stability of evaluation outcomes.

3.4. Model performance evaluation

Upon completion of model training, a rigorous assessment of the predictive performance of the models was conducted using an independent test set. To obtain a comprehensive evaluation of performance, a diverse set of metrics was adopted, including Mean Squared Error (MSE), Mean Absolute Error (MAE), Coefficient of Determination (R^2), and Root Mean Squared Error (RMSE)[59]. Each of these metrics serves a unique purpose in assessing predictive model performance.

We selected these metrics because they enable a multifaceted evaluation of the model's predictive capabilities. MSE and RMSE measure the squared difference between predicted and actual values, penalizing larger prediction errors more heavily. MSE provides a cumulative measure of the squared errors, while RMSE offers a direct measure of the magnitude of the errors, aiding in a more comprehensive understanding of the model's error characteristics[60,61]. MAE, on the other hand, measures the absolute value of prediction errors and is less sensitive to outliers. R^2 provides a measure of the model's explanatory power, reflecting the accuracy of the model's predictions by indicating the proportion of the variance in the dependent variable that is predictable from the independent variables [62].

By incorporating these metrics, we aim to capture the nuances of model performance across different dimensions. The sensitivity of MSE and RMSE to larger errors is crucial for identifying and mitigating significant prediction discrepancies. MAE's equal treatment of all errors allows us to assess the model's performance without being disproportionately influenced by extreme values. Lastly, R^2's holistic view evaluates the overall goodness-of-fit of the model, ensuring that our predictions are not only accurate but also reliable in the context of the variance explained by our model.

Mean Squared Error (MSE) is a metric that effectively quantifies the mean squared difference between the model's predicted values and the actual observations. A lower MSE indicates higher predictive accuracy of the model, as it reflects a closer alignment between the predicted and observed outcomes.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2 \# \tag{2}$$

Where n represents the number of samples,y_i denotes the true value,$\widehat{y}_i$ represents the predicted value.

Root Mean Squared Error (RMSE), as the square root of MSE, provides a measure of the standard deviation of prediction errors, facilitating an intuitive comparison with the units of actual values. RMSE offers a more direct way to assess the magnitude of prediction deviations.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2} \# \tag{3}$$

Where n represents the number of samples,y_i denotes the true value,$\widehat{y}_i$ represents the predicted value.

Mean Absolute Error (MAE) directly reflects the average absolute difference between the model's predicted values and the actual values, eliminating the influence of the sign of errors and thus more genuinely capturing the size of prediction errors.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \widehat{y}_i| \# \tag{4}$$

Where n represents the number of samples,y_i denotes the true value,$\widehat{y}_i$ represents the predicted value.

Coefficient of Determination (R^2), this coefficient profoundly reveals the extent to which the model explains the variability in the data. As the R^2 value approaches 1, it indicates a higher degree of fit between the model and the data[63].

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \overline{y}_i)^2} \# \tag{5}$$

Where n represents the number of samples,y_i denotes the true value,$\widehat{y}_i$ represents the predicted value, $\overline{y}_i$ represents the average of the true values.

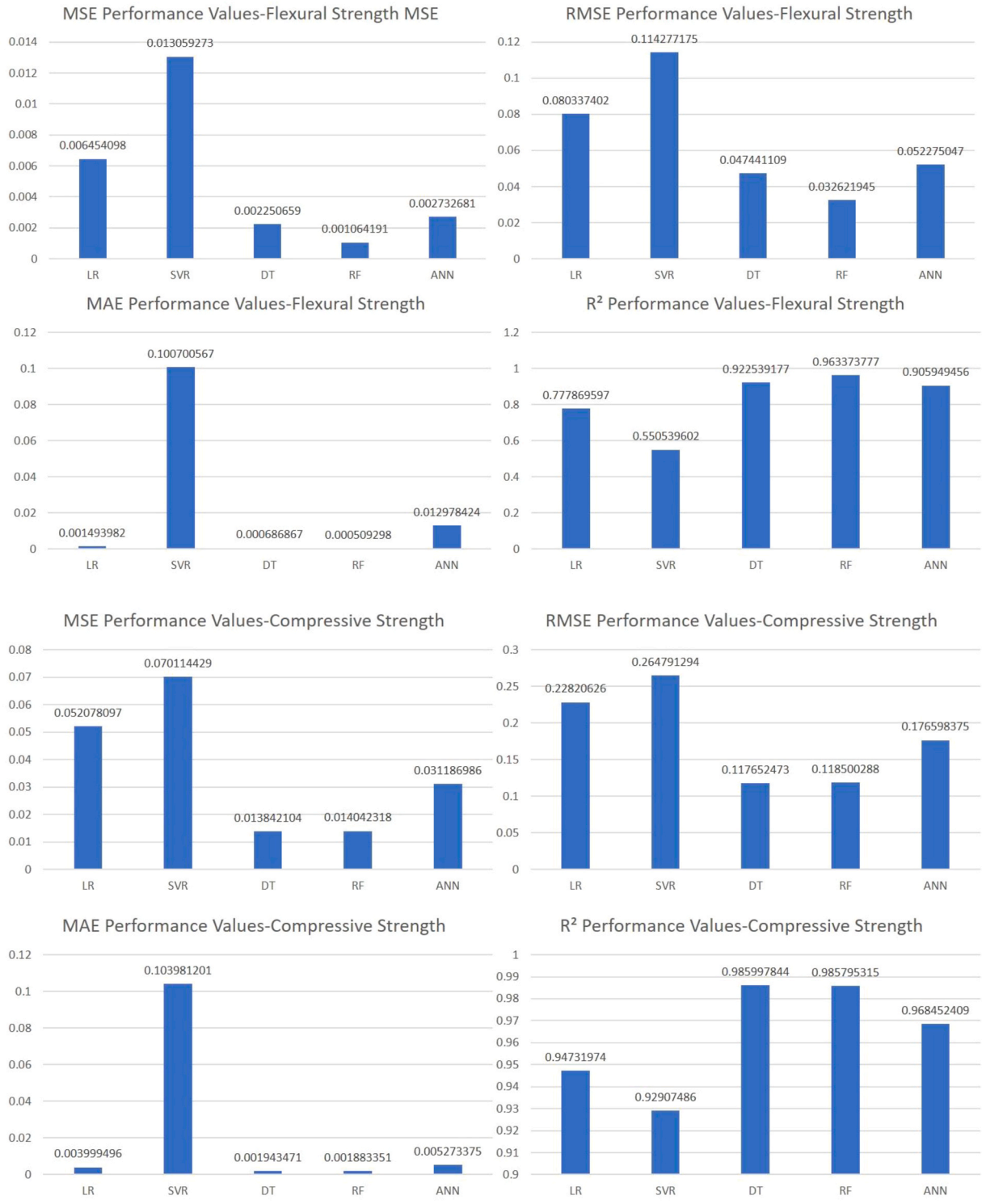


Fig. 5. Evaluation metrics of regression model performance values.

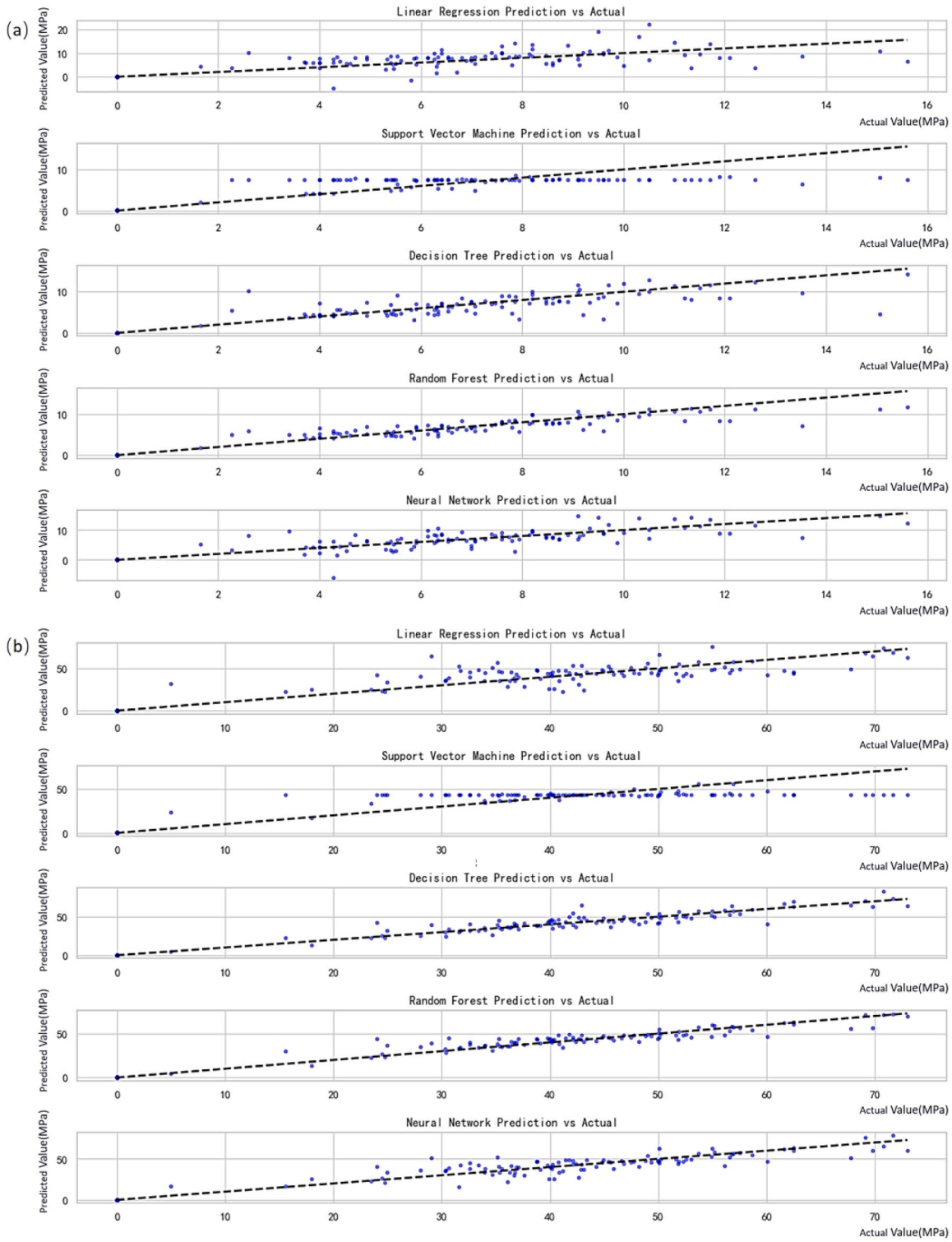


Fig. 6. Scatter plots of differences between predicted and actual values, (a) flexural strength, (b) compressive strength.

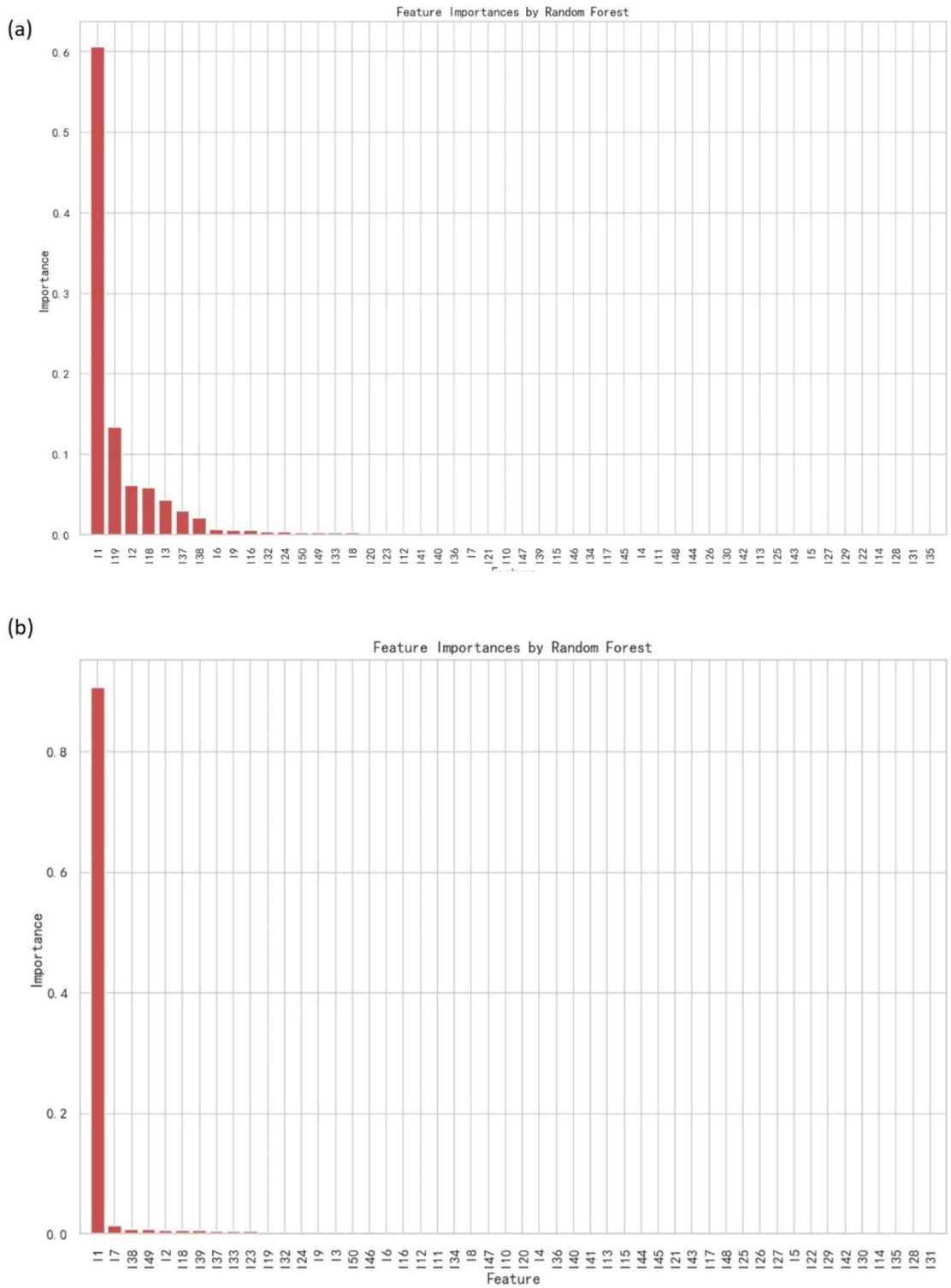


Fig. 7. Random forest sensitivity analysis. (a) flexural strength, (b) compressive strength.

4. Results and discussion

4.1. Prediction results and comparative analysis

Fig. 5 presents the performance metrics for evaluating the regression models. Several machine learning methods showed exceptional and notable performance in the job of predicting the compressive and flexural strengths of 3D-printed concrete. In particular, the Linear Regression model performed exceptionally well when dealing with linear correlations because to its ease of use and effectiveness. It obtained an MSE of 0.006, RMSE of 0.08, MAE of 0.001, and R2 of 0.777 in its prediction of flexural strength. While RMSE (0.228) and MSE (0.052) were somewhat higher for compressive strength prediction, MAE stayed low at 0.003, and R2 reached 0.947. This suggests that linear correlations between specific features and intensities in the data can be effectively captured using linear regression.

However, Support Vector Machine are not the optimal choice for regression problems. In flexural strength prediction, SVM yielded an MSE of 0.013, RMSE of 0.114, MAE of 0.100, and R^2 of 0.55. For compressive strength prediction, the MSE was 0.0701, RMSE was 0.264, MAE was 0.103, and R^2 was 0.929. This result may be attributed to the fact that SVM excel more in classification tasks than regression problems and are sensitive to the scale and feature distribution of data.

In contrast, DT models demonstrated remarkable performance in prediction tasks due to their ability to handle nonlinear relationships and low data preprocessing requirements. For flexural strength prediction, DT achieved an MSE of 0.002, RMSE of 0.047, MAE of 0.0006, and R^2 of 0.922. In compressive strength prediction, the MSE was 0.013, RMSE was 0.117, MAE was 0.001, and R^2 reached an impressive 0.985. Additionally, the ease of understanding and interpretation of Decision Trees enhances the model's interpretability.

The RF model, by integrating multiple DT, further improves the model's stability and generalization ability. In flexural strength prediction, RF achieved an MSE of 0.001, RMSE of 0.032, MAE of 0.0005, and an R^2 of 0.963. For compressive strength prediction, the MSE was 0.014, RMSE was 0.118, MAE was 0.001, and R^2 also reached 0.985. This makes RF a powerful tool for predicting concrete strengths.

While ANN models possess formidable learning capabilities and proficiency in handling complex nonlinear relationships, they demand rigorous data preprocessing and meticulous parameter tuning. In flexural strength prediction, the ANN model exhibited relatively good performance with an MSE of 0.002, RMSE of 0.052, and R^2 of 0.905. However, in compressive strength prediction, its performance slightly declined, with an MSE of 0.031, RMSE of 0.176, and MAE of 0.005, which were higher compared to other models. This underscores the need for more extensive tuning efforts to achieve optimal performance with ANN.

Fig. 6 provides a detailed comparison of the performance of five different machine learning models in predicting the actual versus predicted flexural strengths. Each model's scatter plot visually represents the relationship between predicted and actual values, with the horizontal axis indicating the sample index and the vertical axis representing the flexural strength values. The red horizontal line clearly marks the actual values, while the closely aligned blue dots signify the predictions from each model. Notably, the RF model's predictions align most closely with the actual values, demonstrating its superior predictive performance on this dataset. In contrast, while models such as LR, SVR, and DT also exhibit predictive capability across different sample intervals, their predictions deviate relatively more from the actual values. These scatter plots not only reveal the predictive accuracy of each model within specific sample intervals but also support the assessment and optimization of machine learning models' predictive capabilities through comparative analysis of overall trends.

In conclusion, every algorithm has distinct advantages when it comes to estimating the strength of concrete that has been 3D printed. Because they can identify linear or nonlinear relationships in the data, LR, DT, and RF models perform comparatively better. SVR may not be the best option for such predictions, as evidenced by their comparatively poor performance in regression situations. ANNs require careful parameter adjustment and thorough data pretreatment despite their powerful learning capabilities.

During the trial, the RF model performed best in predicting both compressive and flexural strength. This superiority is ascribed to RF's ensemble learning approach, which combines predictions from several DT to improve prediction accuracy and stability while successfully lowering model variance. LR models, on the other hand, are straightforward and easy to understand, but they could have trouble understanding intricate nonlinear interactions. SVR fared worse than RF in this experiment, although it has advantages for dealing with high-dimensional data and nonlinear situations. Although DT can capture nonlinear relationships, it is unstable and vulnerable to data noise. Due to data volume restrictions or insufficient model complexity modifications, ANN may not have completely utilized their strengths in this experiment, despite their strong representational learning capabilities.

When selecting a model, factors beyond prediction accuracy, such as interpretability, computational complexity, and susceptibility to overfitting, must be considered. For practical applications, choosing a model that balances accuracy with ease of understanding and maintenance may be paramount.

4.2. Sensitivity analysis

One significant benefit of the RF model is its capacity to offer clear explanations of feature relevance. It has proven to function exceptionally well when dealing with data that has intricate nonlinear interactions. Using the RF model, we first determine the importance scores of several elements in this paragraph. We have made a bar chart with the x-axis representing several features and the y-axis showing the associated importance score for each item in order to more clearly and intuitively illustrate the significance of these features. To ensure accuracy and readability, the feature names are generated by processing the temp_df dataframe's column names.

As shown in Fig. 7 (a) The following influencing factors rank from highest to lowest in relevance for flexural strength: cellulose

ether, water-binder ratio, Class I fly ash, 425 cement, belite cement, and polycarboxylate superplasticizer. The most important component affecting flexural strength, according to this study, is the water-binder ratio; Class I fly ash and 425 cement also have rather important places. This implies that flexural strength may be significantly increased by emphasizing changes to the proportions of these components when modifying the formulation of 3D-printed concrete.

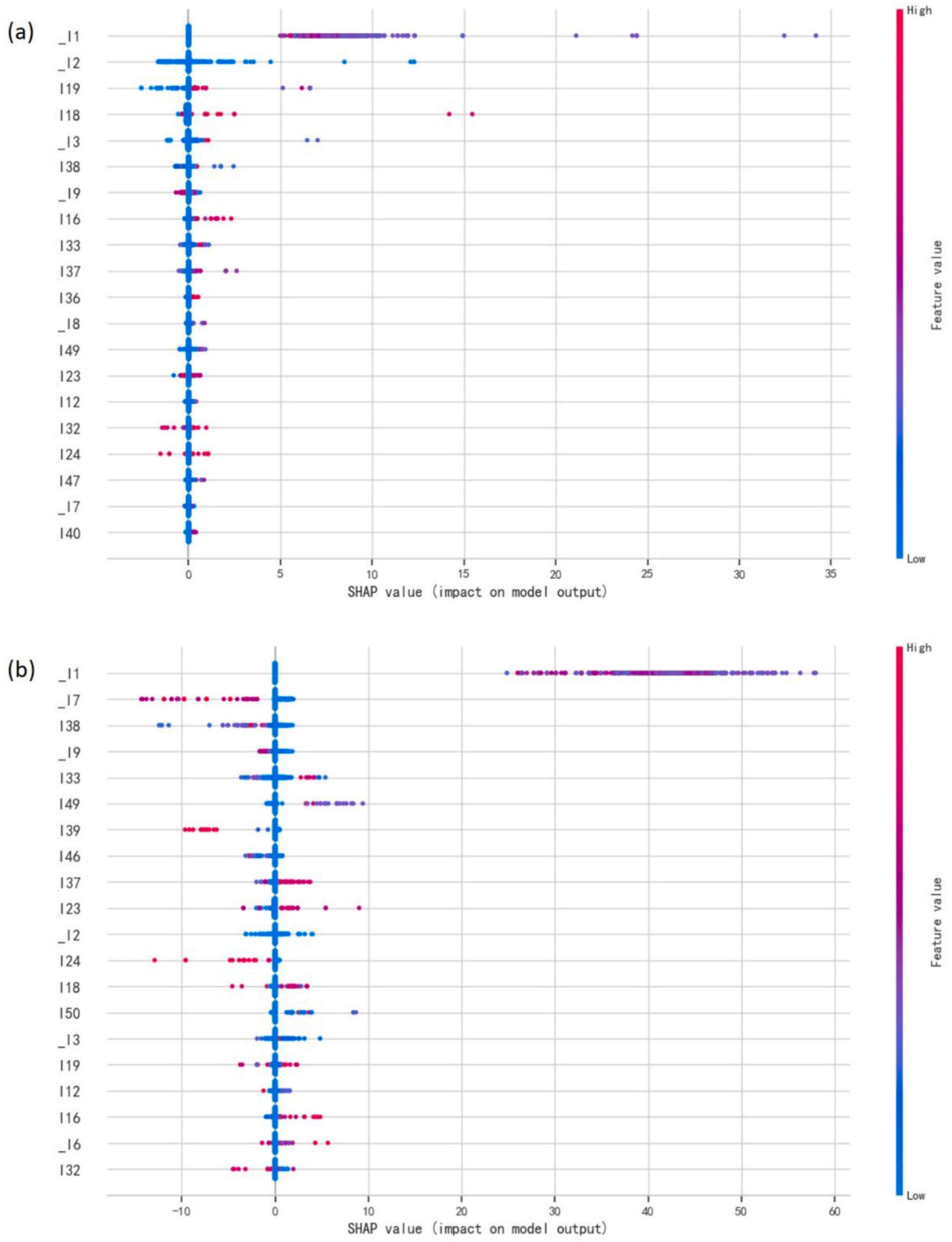


Fig. 8. SHAP analysis, (a) flexural strength, (b) compressive strength.

Water-binder ratio, sulfoaluminate cement, PP fibers, regular Portland cement 425, silica fume, air-entraining agent, polycarboxylate superplasticizer, and tailings are the elements that have the greatest impact on compressive strength, as shown in Fig. 7 (b). Likewise, the ratio of water to binder is found to be the primary determinant of compressive strength. The importance of PP fibers and sulfoaluminate cement is closely followed, demonstrating their significant contributions to improving compressive strength. In order to guide the optimization of 3D-printed concrete formulas to reach better compressive strengths, this study has significant practical implications.

The results indicate that the water-binder ratio plays a paramount role in influencing the mechanical properties of 3D-printed concrete, underscoring the critical importance of meticulously controlling this ratio to ensure optimal concrete performance. The reason is that a decrease in the water-binder ratio leads to an increase in the strength of 3D-printed concrete. Less water means fewer pores, thereby enhancing the density and strength of the 3D-printed concrete. However, an excessively low water-binder ratio may compromise the workability of the 3D-printed concrete, subsequently affecting the 3D printing process. Furthermore, various types of cementitious materials and additives have also exerted notable effects on concrete properties. Specific additives, such as water-reducing agents, air-entraining agents, and fibers, play a significant role in enhancing the mechanical properties of 3D-printed concrete. Water-reducing agents can decrease the amount of water required in the concrete mixture while maintaining the necessary workability, thereby improving the strength and durability of the concrete. Air-entraining agents introduce microbubbles into the concrete, enhancing its mechanical strength, freeze-thaw resistance, and durability. The incorporation of fibers, on the other hand, increases the toughness and crack resistance of the concrete, which is particularly crucial for the long-term performance of 3D-printed concrete structures. These findings provide vital insights for optimizing the formulation of 3D-printed concrete in practical engineering applications, thereby enhancing its mechanical properties and fostering the widespread adoption of 3D printing technology in the construction industry.

4.3. SHAP analysis

This section focuses on model interpretation leveraging SHAP (SHapley Additive exPlanations) technology[64]. For the SHAP analysis, we consider the prediction results from the Random Forest model, given its highest accuracy in predicting the compressive and flexural strengths of 3D-printed concrete. SHAP provides us with a measure of the contribution of each feature to the model's predictions. This helps us identify which input parameters (such as cement content, water-cement ratio, fiber content, etc.) have the most significant impact on the mechanical strength of 3D-printed concrete. Through global visualization, it displays the contribution of each feature for a single prediction or across the entire dataset. These charts allow us to see how the impact of each feature varies with different value ranges, offering insights into the nonlinear relationships captured by the model.

The influence of different input parameters on the anticipated flexural strength inside the RF model is thoroughly examined in Fig. 8 (a). The SHAP values are represented by the horizontal axis, which graphically illustrates how much each parameter affects the model's output. Whereas negative values imply the opposite effect, positive SHAP values show a positive contribution to the forecast. The features are arranged according to their average contribution to the model's prediction outcome (i.e., the sum of absolute SHAP values), whereas the vertical axis lists the specific input parameter symbols. Features that contribute more are usually positioned higher. The redder the color, the more important the feature; the bluer the color, the less important the feature. The red and blue parts behind each feature in the chart are composed of numerous scatter points, with each point representing a sample. Areas with more samples will appear thicker. As evident from the figure, some parameters, such as I1, I2, and I19, exhibit substantial SHAP values, indicating their crucial roles in the model's predictions and significant influence on flexural strength. Conversely, parameters like I15 have smaller SHAP values, suggesting limited impact on the model's output.

Furthermore, the scatter points' slow color change from blue to red highlights the difference in SHAP values, with blue points denoting low influence and red points denoting high SHAP values (i.e., very influential factors). The frequency of positive contribution points for I18, for example, indicates that a rise in the value of this characteristic generally improves the model's prediction. Similar to this, most I1 values are in the positive range, with more red dots than blue, suggesting that an increase in I1 typically results in a higher forecast from the model. The quick identification of crucial parameters in model predictions is made easier by this visual representation. In addition to illustrating the distinct effects of each input parameter on the Random Forest model's flexural strength prediction, the picture serves as an essential guide for further model improvement and parameter modifications. Further model improvements that improve the precision and dependability of mechanical property predictions can be made by exploring these highly significant characteristics in further detail.

Fig. 8 (b) illustrates the distribution of SHAP values, providing an intuitive understanding of the importance of different parameters to the model's prediction results and their directional impact. Analyzing the vertical axis, the feature values of I1, I7, I38, I49, and I39 are sorted in descending order of their contributions, all demonstrating significant impacts. Along the horizontal axis, within the range of −20–60, features such as I1, I49, and I50 exhibit predominantly positive values, indicating they tend to increase the model's predictions. On the other hand, I7 and I38 bring the model's forecasts down into the negative territory. Features like I1, I49, and I37 are darker in color when looking at the point distribution, indicating a larger impact on the anticipated compressive strength. I2 and I3, on the other hand, cluster around zero, suggesting a comparable influence on the prediction results for a large number of samples. Additionally, the overwhelming number of positive contribution points for I1 and I49 emphasizes how raising their values typically improves the model's predictions.

4.4. Factors influencing the accuracy of mechanical strength prediction

The predictive accuracy of the mechanical properties of 3D-printed concrete is intricately linked to multiple synergistic factors, with data quality and quantity being paramount. In this study, nearly 500 sets of experimental data were employed to achieve more precise predictive outcomes. The accuracy, completeness, and consistency of data serve as the foundation for models to discern genuine relationships; any deficiency in data quality may misdirect the model's learning trajectory. In this research, we ensured that each dataset underwent rigorous screening and validation to guarantee high-quality inputs for model training and prediction.

The significance of feature selection and engineering cannot be overlooked. Meticulously selecting features closely associated with the mechanical properties of concrete and applying appropriate preprocessing and transformation techniques, such as feature scaling and normalization, significantly enhance the model's ability to express complex relationships. These feature engineering techniques not only simplify the learning difficulty for models but also bolster their adaptability to new data. Through this approach, we ensure that the model maintains efficient and accurate performance when processing data in practical applications.

Furthermore, model selection and parameter fine-tuning are critical steps in determining predictive accuracy. Each machine learning model possesses unique characteristics that render it more suitable for certain types of datasets. Consequently, selecting the most appropriate model for predicting the mechanical properties of 3D-printed concrete and meticulously tuning its parameters are effective means to enhance predictive precision. For each machine learning model, we conducted parameter tuning experiments, systematically varying key model parameters and documenting model performance under each parameter configuration. For instance, in the random forest model, we adjusted the number and depth of trees; in neural networks, we tested different numbers of layers, neurons, and learning rates. These experiments enabled us to identify the optimal parameter settings and quantify the specific impact of these parameters on model performance. Concurrently, we employed strategies such as cross-validation and regularization to guard against overfitting and underfitting, ensuring that models not only excel on training data but also maintain robust generalization capabilities on new data. This step is crucial for enhancing the model's reliability and practicality.

Through the aforementioned quantitative analysis and performance comparison, we were able to clearly delineate the strengths and weaknesses of each model. We found that linear regression models perform well when data is highly linear, but their predictive capabilities are limited when confronted with complex nonlinear relationships. In contrast, although artificial neural networks excel at capturing complex patterns, they require more data and computational resources, which were not optimal in this study due to the limited dataset. The random forest model, leveraging its exceptional ensemble learning strategy, significantly outperformed other models in predictive performance, achieving a correlation coefficient as high as 0.98 with actual measurements, thereby validating its superiority in complex predictive tasks with small datasets. To comprehensively assess the efficacy of the models, we conducted a comparative analysis between the Random Forest model and advanced deep learning techniques, particularly CNNs. CNNs excel in processing image data related to concrete surface cracks and damage, yet in the context of this study, their requirement for extensive data and computational resources was not optimal due to the limited dataset. Through this comparison, it was observed that the Random Forest model, leveraging its superior ensemble learning strategy, significantly outperformed other models in predictive performance, achieving a correlation coefficient as high as 0.98 with actual measurements, thereby validating its superiority in complex predictive tasks involving small datasets. This comparison not only highlights the unique strengths and limitations of both models in predicting the mechanical properties of concrete but also further substantiates the effectiveness of the Random Forest model in handling structured data and datasets with limited size.

4.5. Random forest model effectiveness analysis

In our analysis, we have underscored the effectiveness of the Random Forest (RF) model in the context of predicting the mechanical properties of 3D-printed concrete, particularly when confronted with a limited dataset. Despite the constraints of limited data, the RF model's ensemble methodology stands out for its ability to effectively prevent overfitting, a common issue in small datasets. This is achieved by leveraging an ensemble approach that synthesizes predictions from multiple decision trees, thereby averaging out the variability and providing stable and accurate predictions.

The high accuracy of the RF model is not just a product of its ensemble nature but also its ability to aggregate predictions from various decision trees, which collectively reduce the impact of errors that might arise from relying on a single tree. This aggregation of predictions leads to a more comprehensive understanding of the data's underlying patterns, enhancing the model's predictive power. Moreover, the RF model's robustness to noise and outliers is a significant advantage. It ensures that anomalies in the data have a minimal effect on the model's performance, allowing for reliable predictions even when the data is not perfectly clean or homogeneous. This robustness is particularly beneficial in the context of 3D-printed concrete, where data can often be subject to variability due to the complex interplay of material properties and printing parameters. As a non-parametric method, the RF model offers the flexibility to handle diverse datasets without presupposing strict relationships between variables. This adaptability is crucial for our study, as it allows the model to identify complex patterns in the data without being confined by rigid assumptions that may not hold true for the nuanced properties of 3D-printed concrete. Furthermore, the RF model's capacity for feature importance assessment is a valuable tool in our research. By evaluating the contribution of each feature to the overall variance reduction across all trees, we can pinpoint the most influential input parameters for predicting the mechanical properties of 3D-printed concrete. This insight aids in the optimization of the model by focusing on the most relevant features, thereby improving prediction accuracy and efficiency. Lastly, the RF model streamlines the data preprocessing stage by naturally accommodating missing values and showing resilience to outliers. This characteristic not only simplifies the preprocessing workflow but also contributes to the overall enhancement of the model's performance and reliability in our analysis. By reducing the susceptibility to outliers and efficiently handling missing data, the RF model

demonstrates its practical utility in scenarios where data preprocessing can be a challenging and time-consuming process.

The Random Forest model's ensemble approach, robustness to noise, non-parametric flexibility, feature importance assessment, and simplified data preprocessing make it an ideal choice for predicting the mechanical properties of 3D-printed concrete, even with a limited dataset. These attributes collectively contribute to the robustness and reliability of our predictive model, aligning with the rigorous standards required for scientific research and practical applications in the field of construction materials.

4.6. Limitations and recommendations for improved predictions

This study employs a machine learning-based prediction approach to offer fresh perspectives on the prediction of mechanical properties in 3D printed concrete. Traditional methods often struggle to fully encapsulate the intricate behavior of 3D printed concrete, but our model has yielded several groundbreaking discoveries. Firstly, utilizing machine learning models, notably the Random Forest model with correlation coefficients ranging between 0.96 and 0.98, we have uncovered the complex nonlinear relationships between the mechanical properties of 3D printed concrete and factors such as its constituent materials and printing parameters. These relationships are frequently challenging to articulate directly through traditional physical models. Moreover, our model has pinpointed the most influential features, including cement type, water-to-binder ratio, and admixtures, that significantly impact mechanical properties, thereby providing a foundation for material design and optimization. Secondly, our research illustrates that by accumulating extensive experimental data and applying machine learning algorithms, we can achieve highly precise predictions of 3D printed concrete's mechanical properties. This data-driven prediction technique not only boosts prediction accuracy and efficiency but also furnishes a novel tool for comprehending the behavior of 3D printed concrete under various conditions. Furthermore, while our machine learning models are data-driven, they harmonize well with the fundamental principles and domain expertise of concrete mechanics. For example, the key features identified by our models, such as cement content and water-to-cement ratio, align with established influencing factors in concrete mechanics. Simultaneously, our model's predictions adhere to the physical laws governing strength, toughness, and other fundamental properties in concrete mechanics. Lastly, our study not only confirms the efficacy of machine learning in predicting the mechanical properties of 3D printed concrete but also paves the way for future research avenues. By exposing the complexity and nonlinear characteristics of 3D printed concrete's behavior, we have established a solid foundation for further refining material design, enhancing printing quality, and improving prediction accuracy.

In this study, we employed five machine learning models, including RF, DT, SVR, LR, and ANN, to predict the mechanical strength of 3D-printed concrete. However, each model exhibited unique limitations in practical applications. The RF model, while capable of comprehensively considering the influence of multiple input parameters on the mechanical strength, is sensitive to parameter settings and prone to overfitting. Additionally, the model's interpretability is relatively weak. To enhance the interpretability of the RF model, we adopted feature importance assessment and SHAP analysis. The DT model is intuitive and easy to understand but tends to suffer from overfitting when dealing with complex materials like 3D-printed concrete, and it requires a large number of training samples to ensure model stability. The SVM model has an advantage in handling nonlinear relationships but is computationally intensive and highly sensitive to the choice of parameters and kernel functions. The LR model is simple and straightforward to implement but is only suitable for capturing linear relationships between input parameters and the mechanical strength of 3D-printed concrete, making it ineffective for modeling complex nonlinear relationships, and it is also sensitive to outliers. The ANN model possesses powerful nonlinear modeling capabilities but involves complex and time-consuming training processes, is prone to overfitting, and, as a black-box model, its decision-making process is difficult to interpret, posing challenges for model application and optimization.

Despite the notable achievements in this study, it is crucial to acknowledge its limitations. One significant bottleneck affecting model prediction performance is the quality of data, which underscores its paramount importance. In the field of 3D printed concrete, data scarcity poses a significant challenge. Due to the relative novelty of the technology and specific testing requirements, obtaining large and diverse datasets is difficult, which may limit the model's ability to learn from a wide range of scenarios and increase the risk of overfitting or underfitting. To overcome this challenge, data preprocessing becomes particularly important. It involves handling missing values to maintain the integrity of the dataset, as well as normalizing and scaling features to ensure that no single feature dominates the model due to its size. Additionally, noise and outliers in the data have a significant impact on model performance. We need to employ relevant techniques to effectively identify and manage these outliers, reducing the noise that may mislead the model. In the context of limited data, feature selection becomes another major challenge. We need to combine domain knowledge with statistical methods to identify the key features that have the greatest impact on the performance of 3D printed concrete, which is crucial for building a model with strong generalization ability. Furthermore, ensuring that the data truly represents the variability and distribution of 3D printed concrete properties is essential. We strive to collect data from different sources and conditions to enhance the representativeness of the dataset. Finally, to maintain data consistency and reliability, we have standardized data collection and preprocessing protocols to ensure that all data points are comparable. In future endeavors, we should prioritize strengthening data quality control and management by adopting more advanced data cleaning and preprocessing techniques to ensure the purity and integrity of data. Regarding model selection, while we have explored five commonly used models, our quest should not cease here. Incorporating a more diverse array of machine learning models, such as deep learning and ensemble learning, and conducting comparative analyses to identify a more fitting prediction scheme, is essential. Furthermore, deeply mining and meticulously constructing features in conjunction with domain knowledge represents a vital direction for enhancing model prediction capabilities. In future research, we plan to adopt automated feature selection and dimensionality reduction techniques, aiming to significantly enhance predictive accuracy. This strategy will not only streamline the dataset dimensions by precisely preserving features closely related to the core issues, thereby reducing the negative impact of noise and irrelevant features and enhancing the model's predictive performance; but it will also effectively reduce computational costs, shorten model training cycles, and conserve resources, which is

particularly important for large-scale data processing, greatly simplifying the cumbersome processes of model training and prediction. Moreover, by refining core features, the technology makes the model results more interpretable, increasing the transparency of the prediction process and facilitating in-depth understanding and validation by experts. Most importantly, the application of dimensionality reduction helps reduce model complexity, effectively avoiding the risk of overfitting, ensuring that the model performs well on training data while also demonstrating good generalization capabilities on new data. Combining domain knowledge to deeply mine and meticulously construct features is another important direction for enhancing the predictive power of models. We will fully leverage the knowledge of domain experts to deeply explore potential features in the data and provide the model with richer and more accurate information inputs. Through meticulous feature engineering, we can enhance the model's predictive ability for the mechanical strength of 3D-printed concrete. Lastly, for optimizing and adjusting model parameters, we can leverage advanced techniques like cross-validation and grid search to automate parameter tuning and achieve precise configurations. This approach not only elevates model prediction accuracy but also shortens the development cycle and reduces costs.

5. Conclusion

By methodically introducing and comparing several machine learning models, this study was able to create a framework for precisely forecasting the flexural and compressive strengths of concrete that has been 3D printed. With a correlation as high as 0.98 with real measurements, experimental results showed that the Random Forest model, with its remarkable ensemble learning strategy, significantly outperformed other models in predictive performance, validating its superiority in challenging prediction tasks. This model demonstrated its dependability in real-world applications by doing exceptionally well on the training set and demonstrating strong generalization abilities on the test set.

This research's main contribution is the efficient use of cutting-edge machine learning algorithms to precisely forecast the mechanical characteristics of 3D-printed concrete, offering a solid scientific foundation for material design and optimization. We can quickly determine the ideal ratios of concrete ingredients by using this predictive framework, which will significantly cut down on the number of trial cycles and resource usage. This not only speeds up the R&D process but also creates a strong basis for the commercialization and industrial use of 3D-printed concrete technology. Furthermore, our mechanical property predictions' high accuracy is essential to guaranteeing the durability, dependability, and structural safety of building structures. This makes it possible to design and build more resilient, eco-friendly, and efficient structures, which greatly advances the innovation and sustainable growth of the construction sector.

Even with this study's noteworthy accomplishments, data control, model variety, and parameter optimization can still be strengthened. For example, the prediction of compressive and flexural strengths was the main emphasis of this work; more intricate performance indicators such interlayer bonding strength, printing parameters, anisotropy, and durability were not thoroughly examined; these will be the subject of our subsequent research.

To improve model performance and data insights, future studies can investigate further advancements in feature engineering and data pretreatment. The development of various methods and parameter optimization are also crucial avenues for the field's advancement. In order to enable our model to forecast more intricate performance metrics such environmental effects, interlayer bonding strength, anisotropy, rheological property, and durability, we intend to conduct additional tests and gather more data.

CRediT authorship contribution statement

Xinxin Wang: Formal analysis, Data curation. **Tao Liu:** Visualization, Resources. **Suping Cui:** Writing – review & editing, Methodology. **Bohao Yang:** Validation, Resources, Investigation. **yonghong zhang:** Writing – original draft, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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